

Decomposition Using Multivalued Attribute (4NF) — DBMS

Definition: A relation is in **Fourth Normal Form (4NF)** if it is in **BCNF** and for every non-trivial **Multivalued Dependency (MVD)** $X \twoheadrightarrow Y$, X is a Super Key.

1. Detailed Technical Explanation

Multivalued Dependency (MVD):

A multivalued dependency $X \twoheadrightarrow Y$ (read as "X multidetermines Y") exists in schema R if the presence of a pair of tuples (t_1, t_2) with $t_1[X] = t_2[X]$ implies that there must also exist tuples t_3 and t_4 combining X, Y , and the remaining attributes $Z = R - (X \cup Y)$.

MVD Symbol: $X \twoheadrightarrow Y$ (X independent multidetermines Y)

Example: 4NF Decomposition

Relation STUDENT_INFO(Student_ID, Mobile_No, Skill)

- A student can have multiple mobile numbers AND multiple independent skills.
- Multivalued Dependencies:**
 - Student_ID $\twoheadrightarrow$ Mobile_No
 - Student_ID $\twoheadrightarrow$ Skill

Data Tuple Redundancy (UNF/BCNF Violation):

Student_ID	Mobile_No	Skill
101	9876543210	Java
101	9876543210	Python
101	9123456789	Java
101	9123456789	Python

(Note: Adding 1 new skill for student 101 requires inserting 2 new rows because Mobile_No and Skill are independent!)

4NF Decomposition Algorithm:

If $X \twoheadrightarrow Y$ violates 4NF in R, decompose R into:

- $R_1 = X \cup Y$
- $R_2 = R - Y$

Decomposed 4NF Tables:

1. **STUDENT_MOBILE** (Student_ID, Mobile_No)

2. **STUDENT_SKILL** (Student_ID, Skill)

2. Core Concepts & Memory Keywords

- **Multivalued Dependency (MVD):** Independence between two multi-valued attributes associated with the same key.
 - **Spurious Combinations:** Combinatorial explosion of duplicate tuples caused by MVDs.
 - **4NF Condition:** Every non-trivial MVD $X \twoheadrightarrow Y$ must have X as a super key.
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3. Must-Write Points for Exams

- 4NF handles independent multi-valued attributes that BCNF cannot resolve.
 - MVDs occur when two independent 1-to-many relationships are combined in a single relation.
 - Decomposing MVDs into separate 2-column tables eliminates combinatorial row insertion anomalies.
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4. Quick Recall Flow

BCNF Table $\rightarrow$ Identify Independent Multi-Valued Attributes ($X \twoheadrightarrow Y$) $\rightarrow$ Split into $R_1(X, Y)$ & $R_2(X, Z)$ $\rightarrow$ 4NF Achieved